



Capacitive Foot Pressure Mat (CFPM)

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MAKE WAVES.

Abstract

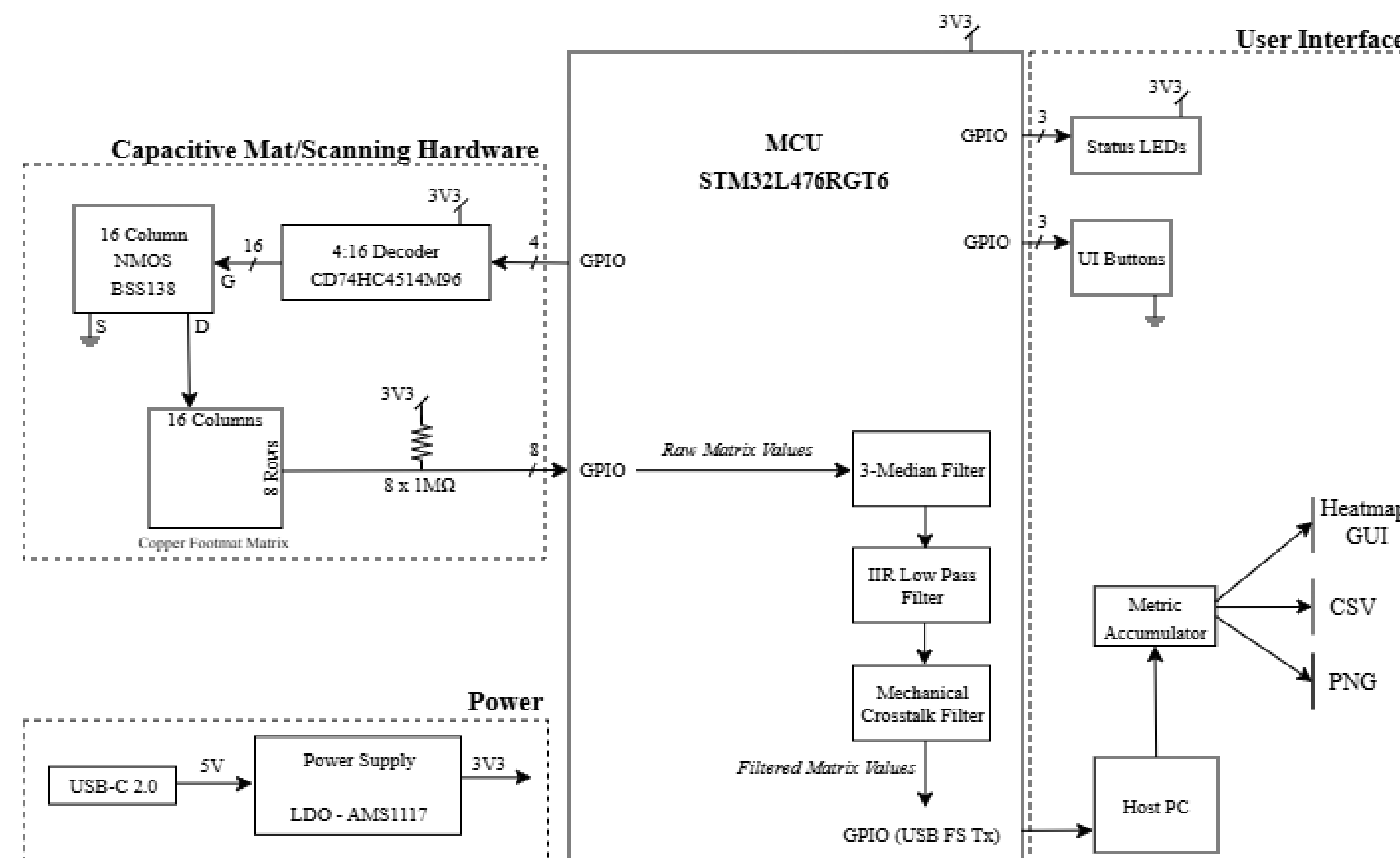
During the 2023-2024 school year, I (Reese) had suffered from two herniated discs and had to have back surgery. During my rehabilitation/physical therapy process, I had to monitor my progress in correcting the pressure distribution of my feet.

This system aims to achieve the best solution for tracking said progress while using the cheapest possible materials.

A low-power STM32 microcontroller applies a charging and scanning sequence (*drawn up by Prof. Lund*) to a physical copper matrix which the user stands on, forming capacitors at each copper intersection point. Values of capacitance at each intersection are determined by how much pressure the user applies at each intersection (more weight leads to higher compression of the intersection's dielectric, bringing the top and bottom capacitor plates closer together, thus more capacitance).

Data frames encompassing all intersection values are passed through digital filtering and then transmitted over a USB-C 2.0 A-B cable to a host PC for real-time visualization and record keeping across different user sessions. (All pictured right)

Methods and Materials



The following filters (applied to each full frame of matrix data) were implemented sequentially through empirical testing and trial and error:

- 3-Sample Median Filter
 - Rejects single-frame transient spikes caused by EMI interference picked up by the high impedance row traces
- IIR Low Pass Filter
 - Smooths continuous frame-to-frame jitter which lasts longer than a single-frame spike
- Mechanical Crosstalk Filter
 - Mitigates mechanical crosstalk caused by the rigidity of the backing which the user stands on (i.e. when one point on the top backing is pushed down, the region around it is slightly pushed down as well)

Conclusion

Significance

- Data output is accurate enough to be applied to a vast myriad of physical therapy or sport applications

Hardware Improvements

- Portable Insole Version
- Bluetooth
- Force Sensing Resistors (A/D conversion)

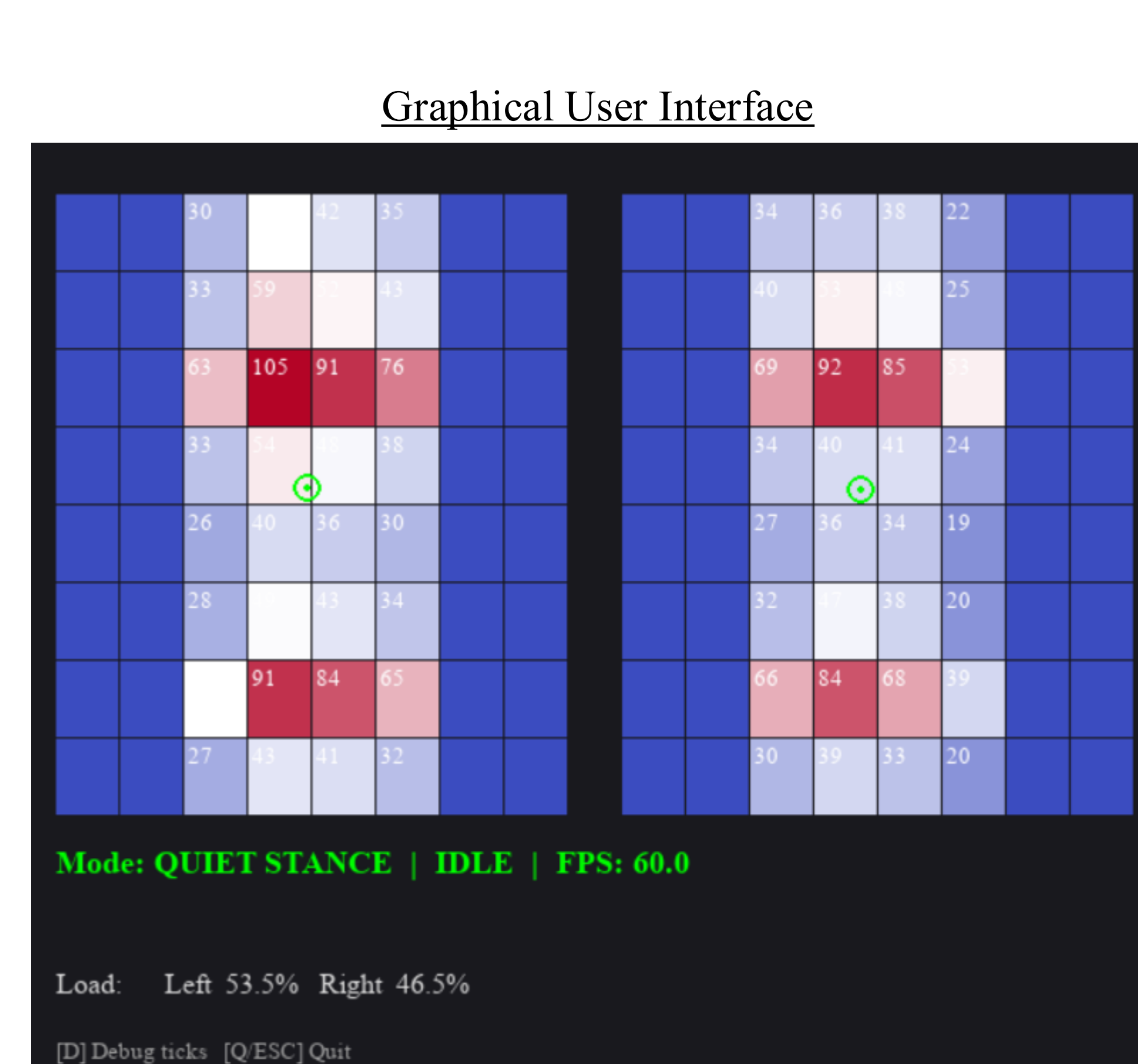
Software Improvements

- Real-Time Health Metric Display Phone App
- Center of Mass Calculation

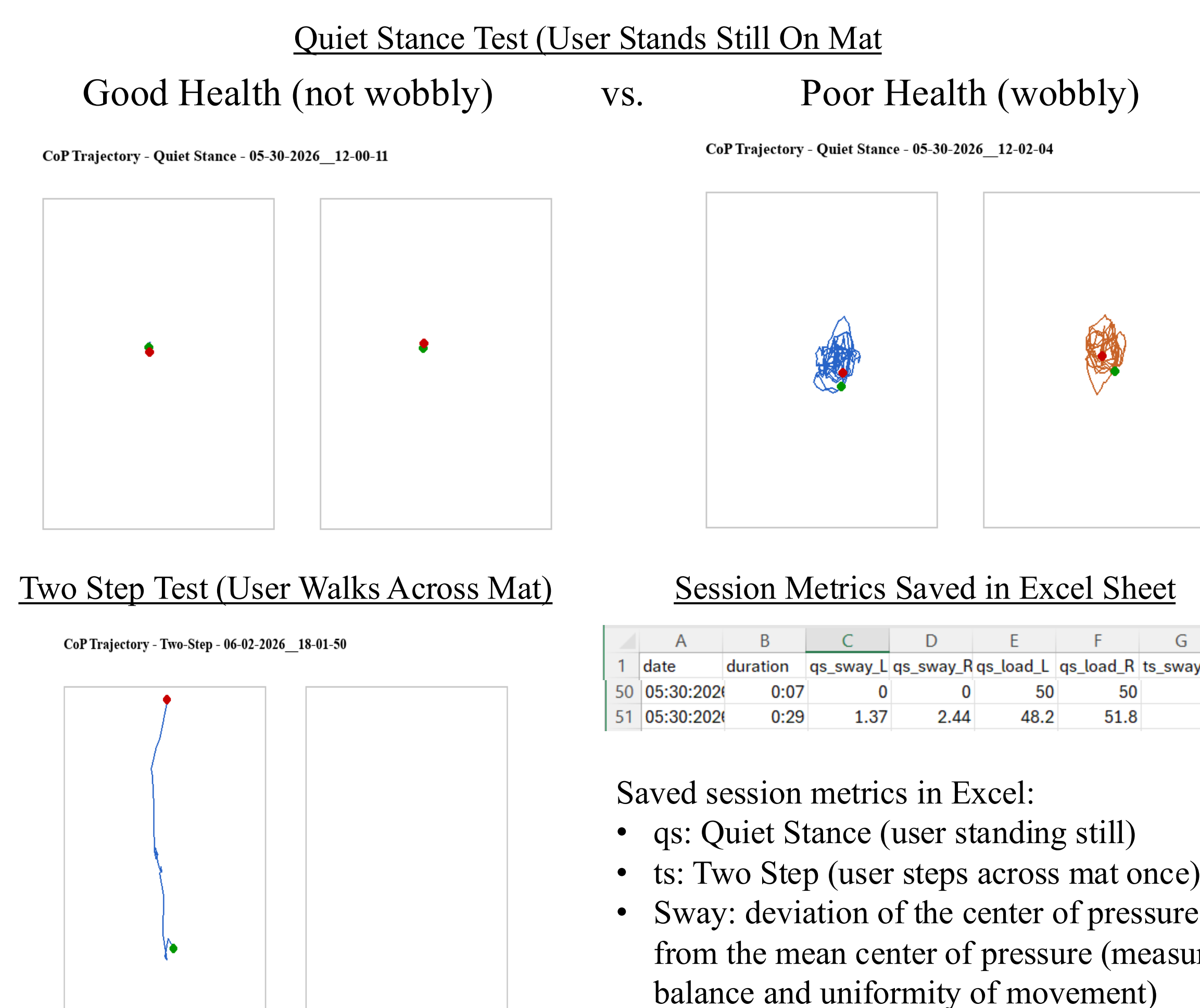
Acknowledgements

Professor Lund, thank you for all your help in hardware development.

Results



Result: Successful real time display of the user's foot pressure (center of pressure) is the green dot).



Result: Accurate recording of the path of travel of the user's center of pressure during standing and walking.

References

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